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*****
*
*      STAAD.Pro
*      Version 2007   Build 04
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date=   DEC 2, 2011
*      Time=   17: 2:55
*
*      USER ID:
*****
```

```
1. STAAD SPACE DXF IMPORT OF PR421-PR425.DXF
INPUT FILE: 20111127 PR421-PR425 RF.STD
2. START JOB INFORMATION
3. ENGINEER DATE 27-OCT-11
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 1 10.0004 0 -9.91821E-005; 2 10.0004 7.81982 -9.91821E-005
9. 3 0.000398636 5.19971 -9.91821E-005; 4 10.0004 5.19971 -9.91821E-005
10. 5 0.000398636 0 -9.91821E-005; 6 0.000398636 7.81982 -9.91821E-005
11. 7 0.000398636 5.19971 -9.1001; 8 10.0004 5.19971 -9.1001
12. 9 0.000398636 0 -9.1001; 10 0.000398636 7.81982 -9.1001
13. 11 10.0004 7.81982 -9.1001; 12 10.0004 0 -9.1001
14. 13 20.0004 5.19971 -9.91821E-005; 14 20.0004 5.19971 -9.1001
15. 15 20.0004 0 -9.91821E-005; 16 20.0004 7.81982 -9.91821E-005
16. 17 20.0004 0 -9.1001; 18 20.0004 7.81982 -9.1001
17. 19 33.0004 7.81982 -9.91821E-005; 20 33.0004 7.81982 -9.1001
18. 21 33.0004 0 -9.1001; 22 33.0004 0 -9.91821E-005
19. 23 33.0004 5.19971 -9.91821E-005; 24 33.0004 5.19971 -9.1001
20. 25 39.0004 0 -9.91821E-005; 26 39.0004 7.81982 -9.91821E-005
21. 27 39.0004 0 -9.1001; 28 39.0004 7.81982 -9.1001; 29 39.0004 5.19971 -9.1001
22. 30 39.0004 5.19971 -9.91821E-005; 31 15.0004 5.19971 -9.91821E-005
23. 32 15.0004 5.19971 -9.1001; 33 5.0004 5.19971 -9.91821E-005
24. 34 5.0004 5.19971 -9.1001; 35 5.0004 7.81982 -9.91821E-005
25. 36 5.0004 7.81982 -9.1001; 37 26.5004 5.19971 -9.91821E-005
26. 38 26.5004 5.19971 -9.1001; 39 26.5004 7.81982 -9.91821E-005
27. 40 26.5004 7.81982 -9.1001; 41 15.0004 7.81982 -9.91821E-005
28. 42 15.0004 7.81982 -9.1001; 43 0.000398636 5.19971 -4.5501
29. 44 0.000398636 7.81982 -4.5501; 45 5.0004 5.19971 -4.5501
30. 46 5.0004 7.81982 -4.5501; 47 10.0004 5.19971 -4.5501
31. 48 10.0004 7.81982 -4.5501; 49 23.2504 7.81982 -9.91821E-005
32. 50 23.2504 5.19971 -9.91821E-005; 51 23.2504 7.81982 -9.1001
33. 52 23.2504 5.19971 -9.1001; 53 29.7504 7.81982 -9.1001
34. 54 29.7504 5.19971 -9.1001; 55 29.7504 5.19971 -9.91821E-005
35. 56 29.7504 7.81982 -9.91821E-005; 57 17.5004 7.81982 -9.1001
36. 58 17.5004 5.19971 -9.1001; 59 12.5004 7.81982 -9.1001
37. 60 12.5004 5.19971 -9.1001; 61 17.5004 7.81982 -9.91821E-005
38. 62 17.5004 5.19971 -9.91821E-005; 63 12.5004 7.81982 -9.91821E-005
39. 64 12.5004 5.19971 -9.91821E-005
40. MEMBER INCIDENCES
```

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41. 1 1 4; 2 5 3; 3 9 7; 4 12 8; 5 15 13; 6 17 14; 7 21 24; 8 22 23; 9 25 30
42. 10 27 29; 11 4 2; 12 3 6; 13 7 10; 14 8 11; 15 13 16; 16 14 18; 17 24 20
43. 18 23 19; 19 30 26; 20 29 28; 21 6 44; 22 3 43; 23 2 48; 24 13 14; 25 19 20
44. 26 16 18; 27 4 47; 28 23 24; 29 31 32; 30 33 45; 31 35 46; 32 37 38; 33 30 29
45. 34 26 28; 35 40 39; 36 3 33; 37 6 35; 38 7 34; 39 10 36; 40 8 60; 41 4 64
46. 42 16 49; 43 13 50; 44 18 51; 45 14 52; 46 20 28; 47 24 29; 48 23 30; 49 19 26
47. 50 33 4; 51 35 2; 52 34 8; 53 36 11; 54 32 58; 55 11 59; 56 31 62; 57 2 63
48. 58 37 55; 59 38 54; 60 39 56; 61 40 53; 62 41 61; 63 42 57; 64 42 41; 65 43 7
49. 66 44 10; 67 45 34; 68 46 36; 69 44 46; 70 43 45; 83 7 45; 85 3 45; 119 43 34
50. 120 43 33; 125 10 46; 126 6 46; 127 44 36; 128 44 35; 141 47 8; 142 45 47
51. 143 48 11; 144 46 48; 145 34 47; 146 33 47; 147 45 8; 148 45 4; 149 36 48
52. 150 35 48; 151 46 11; 152 46 2; 153 4 15; 154 13 1; 155 8 17; 156 14 12
53. 157 23 25; 158 30 22; 159 24 27; 160 29 21; 161 24 30; 162 29 23; 163 20 26
54. 164 28 19; 169 39 37; 170 40 38; 171 49 39; 172 50 37; 173 51 40; 174 52 38
55. 175 53 20; 176 54 24; 177 55 23; 178 56 19; 179 51 52; 180 53 54; 181 49 50
56. 182 56 55; 183 18 52; 184 51 38; 185 53 38; 186 20 54; 187 16 50; 188 49 37
57. 189 56 37; 190 19 55; 191 7 5; 192 3 9; 193 41 31; 194 42 32; 195 57 18
58. 196 58 14; 197 59 42; 198 60 32; 199 61 16; 200 62 13; 201 63 41; 202 64 31
59. 203 59 60; 204 57 58; 205 61 62; 206 63 64; 207 18 58; 208 57 32; 209 59 32
60. 210 11 60; 211 16 62; 212 61 31; 213 63 31; 214 2 64
61. DEFINE MATERIAL START
62. ISOTROPIC STEEL
63. E 2.05E+008
64. POISSON 0.3
65. DENSITY 76.8195
66. ALPHA 1.2E-005
67. DAMP 0.03
68. END DEFINE MATERIAL
69. MEMBER PROPERTY JAPANESE
70. 23 25 26 34 35 64 143 TABLE ST H400X400X13
71. 83 85 119 120 125 TO 128 145 TO 164 191 192 TABLE ST C150X75X6.5
72. 29 32 TABLE ST H400X400X13
73. 2 3 5 TO 10 12 13 15 TO 20 TABLE ST H400X400X13
74. 1 4 11 14 TABLE ST H414X405X18
75. 21 30 31 38 TO 49 52 TO 63 66 TO 68 171 TO 178 195 TO 202 TABLE ST H390X300X10
76. 169 170 179 TO 190 193 194 203 TO 214 TABLE ST H150X150X7
77. 22 27 33 36 37 50 51 65 70 141 142 TABLE ST H900X300X16
78. 24 28 TABLE ST H414X405X18
79. 69 144 TABLE ST H414X405X18
80. CONSTANTS
81. BETA 90 MEMB 1 TO 20
82. MATERIAL STEEL ALL
83. SUPPORTS
84. 1 5 9 12 15 17 21 22 25 27 PINNED
85. MEMBER TENSION
86. 83 85 119 120 125 TO 128 145 TO 164 191 192
87. MEMBER RELEASE
88. 35 64 START MY MZ
89. 35 64 END MY MZ
90. 69 START MY MZ
91. 144 END MY MZ
92. 31 68 START MY MZ
93. 31 68 END MY MZ
94. 29 32 START MY MZ
95. 29 32 END MY MZ
96. 70 START MY MZ
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97. 142 END MY MZ
98. 30 67 START MY MZ
99. 30 67 END MY MZ
100. 46 TO 49 START MY MZ
101. 46 TO 49 END MY MZ
102. 42 TO 45 START MY MZ
103. 175 TO 178 END MY MZ
104. 40 41 55 57 START MY MZ
105. 195 196 199 200 END MY MZ
106. 38 39 START MY MZ
107. 52 53 END MY MZ
108. 36 37 START MY MZ
109. 50 51 END MY MZ
110. MEMBER TRUSS
111. 169 170 179 TO 190
112. MEMBER TRUSS
113. 193 194 203 TO 214
114. LOAD 1 LOADTYPE NONE TITLE DL
115. SELFWEIGHT Y -1
116. MEMBER LOAD
117. 36 50 UNI GY -75.32
118. 27 141 UNI GY -134.15
119. 29 30 32 67 UNI GY -44.76
120. 24 28 UNI GY -89.49
121. 33 UNI GY -74.51
122. 70 UNI GY -73.91
123. 37 51 UNI GY -44.78
124. 23 143 UNI GY -66.5
125. 31 35 64 68 UNI GY -29.24
126. 25 26 UNI GY -48.6
127. 34 UNI GY -39.45
128. 69 UNI GY -22.05
129. LOAD 2 LOADTYPE NONE TITLE FL
130. MEMBER LOAD
131. 36 50 UNI GZ -7.53
132. 27 141 UNI GX -13.42
133. 29 30 32 67 UNI GX -4.48
134. 24 28 UNI GX -8.95
135. 33 UNI GX -7.45
136. 70 UNI GZ -20.12
137. 37 51 UNI GZ -3.74
138. 23 143 UNI GX -5.92
139. 31 35 64 68 UNI GX -1.98
140. 25 26 UNI GX -3.95
141. 34 UNI GX -3.29
142. 69 UNI GZ -3.26
143. LOAD 3 LOADTYPE NONE TITLE WLX
144. MEMBER LOAD
145. 5 TO 35 64 TO 68 141 143 UNI GX -0.36
146. 4 UNI GX -0.36 4.19971 5.19971
147. 4 UNI GX -0.36 0 4.19971
148. 1 UNI GX -0.36 4.19971 5.19971
149. 1 UNI GX -0.36 0 4.19971
150. 2 UNI GX -0.36 4.19971 5.19971
151. 2 UNI GX -0.36 0 4.19971
152. 3 UNI GX -0.36 4.19971 5.19971

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153. 3 UNI GX -0.36 0 4.19971
154. LOAD 4 LOADTYPE NONE TITLE WLZ
155. MEMBER LOAD
156. 5 TO 20 UNI GZ -0.36
157. 4 UNI GZ -0.36 4.19971 5.19971
158. 4 UNI GZ -0.36 0 4.19971
159. 1 UNI GZ -0.36 4.19971 5.19971
160. 1 UNI GZ -0.36 0 4.19971
161. 2 UNI GZ -0.36 4.19971 5.19971
162. 2 UNI GZ -0.36 0 4.19971
163. 3 UNI GZ -0.36 4.19971 5.19971
164. 3 UNI GZ -0.36 0 4.19971
165. 5 TO 70 83 85 119 120 125 TO 128 141 TO 164 171 TO 178 195 TO 202 UNI GZ -0.36
166. 4 UNI GZ -0.36 4.19971 5.19971
167. 4 UNI GZ -0.36 0 4.19971
168. 1 UNI GZ -0.36 4.19971 5.19971
169. 1 UNI GZ -0.36 0 4.19971
170. 2 UNI GZ -0.36 4.19971 5.19971
171. 2 UNI GZ -0.36 0 4.19971
172. 3 UNI GZ -0.36 4.19971 5.19971
173. 3 UNI GZ -0.36 0 4.19971
174. 153 154 157 TO 160 UNI GZ -0.13
175. LOAD 5 LOADTYPE NONE TITLE NLX
176. SELFWEIGHT X -0.015
177. MEMBER LOAD
178. 50 UNI GX -1.13 3 5
179. 50 UNI GX -1.13 0 3
180. 36 UNI GX -1.13 2 5
181. 36 UNI GX -1.13 0 2
182. 141 UNI GX -2 2.55 4.55
183. 141 UNI GX -2 0 2.55
184. 27 UNI GX -2 2 4.55
185. 27 UNI GX -2 0 2
186. 29 30 32 67 UNI GX -0.67
187. 24 28 UNI GX -1.34
188. 33 UNI GX -1.12
189. 70 UNI GX -1.11
190. 37 51 UNI GX -0.67
191. 23 143 UNI GX -0.97
192. 31 35 64 68 UNI GX -0.44
193. 25 26 UNI GX -0.73
194. 34 UNI GX -0.59
195. 69 UNI GX -0.33
196. LOAD 6 LOADTYPE NONE TITLE NLZ
197. SELFWEIGHT Z -0.015
198. MEMBER LOAD
199. 50 UNI GZ -1.13 3 5
200. 50 UNI GZ -1.13 0 3
201. 36 UNI GZ -1.13 2 5
202. 36 UNI GZ -1.13 0 2
203. 141 UNI GZ -2 2.55 4.55
204. 141 UNI GZ -2 0 2.55
205. 27 UNI GZ -2 2 4.55
206. 27 UNI GZ -2 0 2
207. 29 30 32 67 UNI GZ -0.67
208. 24 28 UNI GZ -1.34

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209. 33 UNI GZ -1.12
210. 70 UNI GZ -1.11
211. 37 51 UNI GZ -0.67
212. 23 143 UNI GZ -0.97
213. 31 35 64 68 UNI GZ -0.44
214. 25 26 UNI GZ -0.73
215. 34 UNI GZ -0.59
216. 69 UNI GZ -0.33
217. LOAD COMB 7 1.0DL+1.0FL
218. 1 1.0 2 1.0
219. LOAD COMB 8 1.0DL+1.0FL+1.0WLX
220. 1 1.0 2 1.0 3 1.0
221. LOAD COMB 9 1.0DL+1.0FL+1.0WLZ
222. 1 1.0 2 1.0 4 1.0
223. LOAD COMB 10 1.0DL+1.0FL+1.0NLX
224. 1 1.0 2 1.0 5 1.0
225. LOAD COMB 11 1.0DL+1.0FL+1.0NLZ
226. 1 1.0 2 1.0 6 1.0
227. LOAD COMB 12 1.0DL+1.4FL
228. 1 1.0 2 1.4
229. LOAD COMB 13 1.4DL+1.4FL
230. 1 1.4 2 1.4
231. LOAD COMB 14 1.0DL+1.4FL+1.4WLX
232. 1 1.0 2 1.4 3 1.4
233. LOAD COMB 15 1.0DL+1.4FL+1.4WLZ
234. 1 1.0 2 1.4 4 1.4
235. LOAD COMB 16 1.0DL+1.4FL+1.4NLX
236. 1 1.0 2 1.4 5 1.4
237. LOAD COMB 17 1.0DL+1.4FL+1.4NLZ
238. 1 1.0 2 1.4 6 1.4
239. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 64/ 148/ 10

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 354

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 1

**START ITERATION NO. 2

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**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 2

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 3

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 4

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 5

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 6

240. LOAD LIST 7 TO 11
241. PRINT JOINT DISPLACEMENTS ALL